

$$A^{in} = \begin{bmatrix} 0 & 1 & \dots \\ \vdots & \ddots & \vdots \\ 1 & 1 & \dots \end{bmatrix}_{N_{CT} \times N_{mets}} \quad \text{Diet, } D = \begin{bmatrix} D_1 \\ D_2 \\ \vdots \end{bmatrix}_{N_{mets} \times 1}$$

$$\text{Celltype proportion, } \chi = \begin{bmatrix} p_1 \\ p_2 \\ \vdots \end{bmatrix}_{N_{CT} \times 1}$$

$$\text{Intake, } I = (A^{in} \ D) \odot \chi$$

$$= \left(\begin{bmatrix} 0 & 1 & \dots \\ \vdots & \ddots & \vdots \\ 1 & 1 & \dots \end{bmatrix}_{N_{CT} \times N_{mets}} \begin{bmatrix} D_1 \\ D_2 \\ \vdots \end{bmatrix}_{N_{mets} \times 1} \right) \odot \begin{bmatrix} p_1 \\ p_2 \\ \vdots \end{bmatrix}_{N_{CT} \times 1}$$

$$= \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \vdots \end{bmatrix}_{N_{CT} \times 1}, \text{ where } \sigma_i = p_i \sum_{j=1}^{N_{mets}} A_{i,j}^{in} D_j$$

$$A^{out} = \begin{bmatrix} 0 & 1 & \cdots \\ \vdots & \ddots & \vdots \\ 1 & 1 & \cdots \end{bmatrix}_{N_{CT} \times N_{mets}}$$

Inverse out-degree, $\Omega = \begin{bmatrix} \omega_1^{-1} \\ \omega_2^{-1} \\ \vdots \end{bmatrix}_{N_{CT} \times 1}$

Byproduct fraction, $f = 0.5$

Output, $S = A^{out} \odot (f \ \Omega)$

$$= \begin{bmatrix} 0 & \frac{f}{\omega'_{1,2}} & \cdots \\ \vdots & \ddots & \vdots \\ \frac{f}{\omega'_{N_{CT},1}} & \frac{f}{\omega'_{N_{CT},2}} & \cdots \end{bmatrix}_{N_{CT} \times N_{mets}}, \text{ where } \frac{1}{\omega'_{i,j}} = \frac{1}{\omega_i} A^{out}_{i,j}$$

Secreted metabolome part, $M^s = S^T I$

$$= \begin{bmatrix} 0 & \dots & \frac{f}{\omega'_{N_{CT}, 1}} \\ \frac{f}{\omega'_{1,2}} & \dots & \frac{f}{\omega'_{N_{CT},2}} \\ \vdots & \dots & \vdots \end{bmatrix}_{N_{mets} \times N_{CT}} \begin{bmatrix} \sigma_1 \\ \sigma_2 \\ \vdots \end{bmatrix}_{N_{CT} \times 1}$$

$$= \begin{bmatrix} \sum_{i=1}^{N_{CT}} \sigma_i \frac{f}{\omega'_{i, 1}} \\ \sum_{i=1}^{N_{CT}} \sigma_i \frac{f}{\omega'_{i, 2}} \\ \vdots \end{bmatrix}_{N_{mets} \times 1}$$

Net predicted metabolome, $M^{pred} = M^{unused} + M^s$,

where $\forall k \in \{1, 2, \dots, N_{mets}\}$, $M_k^{unused} = D_k \forall \sum_{i=1}^{N_{CT}} \frac{f}{\omega'_{i,k}} = 0$

and 0 otherwise